

**ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)**

Mid Term-I (Session: 2025-26)

Roll No.:

Course Name: B. Tech.

Subject Name: Environment and Sustainability

Subject Code: 25AS103

Max. Marks: 15

Time: 75 Minutes

Semester: I

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Maximum Knowledge Level: K2)				
Q. No.	Question	CO	KL	PI
1. Attempt All Questions (1 x 5)				
(a)	Enlist and explain briefly the components of an ecosystem.	CO1	K1	9.1.1
(b)	Write any two impacts of urbanization on environment.	CO1	K1	8.2.2
(c)	Explain the pillars of Sustainable Development.	CO1	K1	7.2.2
(d)	Give any two points of importance of Urban mining.	CO2	K1	8.2.2
(e)	Differentiate between bioleaching and phytoremediation.	CO2	K1	8.1.1
SECTION-B (Minimum Knowledge Level: K2)				
2. Attempt Any two Question (Same K Levels) (2 x 2)				
(a)	Explain SDG-13 and write about the measures taken in India related to it.	CO1	K2	7.2.2
(b)	How would you interpret the causes which led to Minamata disease?	CO1	K2	7.2.1
(c)	Based on your understanding, write the challenges in the path of Sustainable Development.	CO1	K2	7.2.2
SECTION-C (Minimum Knowledge Level: K3)				
3. Attempt Any one Question (Same K Levels) (3 x 1)				
(a)	Explain with an example how you can make use of DPSIR framework to assess the impacts of any industry?	CO1	K3	2.4.2
(b)	Discuss the concept of IPAT and explain how it helps in understanding the impact of urbanization?	CO1	K3	2.4.2
4. Attempt Any one Question (Same K Levels) (3 x 1)				
(a)	What will be the scenario in our country India when ecological footprint will exceed the bio-capacity?	CO2	K3	4.3.4
(b)	Explain with diagram the working of Sewage Treatment Plant. Discuss how smart technology can enhance its efficiency.	CO2	K3	5.2.1

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Mid Term-II (Session: 2025-26)

Roll No.:

Course Name: B.Tech
Subject Name: Environment and Sustainability
Subject Code: 25AS103

Max. Marks: 15
Time: 75 Minutes
Semester: I

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Maximum Knowledge Level: K2)				
Q. No.	Question	CO	KL	PI
1. Attempt All Questions (1 x 5)				
(a)	How does fossil fuel dependency contribute to environmental damage?	CO3	K2	9.1.1
(b)	What do you mean by decentralized energy system?	CO3	K1	12.1.1
(c)	Enlist the criteria pollutants required to evaluate AQI.	CO3	K2	9.1.1
(d)	Enumerate the sustainability concern of blockchain.	CO2	K2	2.1.1
(e)	Define the terms deforestation and reforestation.	CO2	K1	8.2.2
SECTION-B (Minimum Knowledge Level: K2)				
2. Attempt Any two Question (Same K Levels) (2 x 2)				
(a)	Explain how green hydrogen differs from grey hydrogen in terms of production and environmental impact?	CO3	K2	2.4.4
(b)	Describe how EVs contribute to lowering greenhouse gas emissions?	CO3	K2	2.4.3
(c)	How does the large-scale biofuel production can lead to an ecological trade-off?	CO3	K2	3.1.1
SECTION-C (Minimum Knowledge Level: K3)				
3. Attempt Any one Question (Same K Levels) (3 x 1)				
(a)	How can adopting cloud-based services minimize hardware requirements and promote green computing?	CO3	K3	1.4.1
(b)	How can IoT sensors be applied to enhance resource use efficiency in modern agriculture?	CO3	K3	2.2.4
4. Attempt Any one Question (Same K Levels) (3 x 1)				
(a)	In what ways does climate change threaten the resilience of an ecosystem?	CO2	K3	4.3.1
(b)	Explain, with an example, how ISWM ensures sustainability compared to SWM?	CO2	K3	4.3.3

CO Course Outcomes mapped with respective question
KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)
K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Pre - End Term Exam (Session: 2025-26)

Admission No.:

Course Name: B.Tech.

Subject Name: Environment and Sustainability

Subject Code: 25AS103

Max. Marks: 20

Time: 90 Minutes

Semester: I

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Knowledge Level: K2 & K3)				
Q. No.	Question	CO	KL	PI
Attempt any one part from each question		(3 Marks x 4 Questions = 12 Marks)		
1.(a)	Enlist the benefits of ISO 14001 and explain its key elements.	CO4	K2	6.1.1
1.(b)	Define LCA. Explain the components of Life Cycle Assessment.	CO4	K2	7.1.1
2.(a)	What are Certified Emission Reductions (CERs)? Explain the goal of Clean Development Mechanism.	CO4	K2	6.1.1
2.(b)	'The usage of nanoparticles in various fields has led to some environmental degradation'. Justify the statement with examples.	CO4	K3	2.2.3
3.(a)	Explain the agenda of Convention of Biological Diversity (CBD), Kyoto Protocol and Paris Agreement.	CO5	K2	12.2.1
3.(b)	Depict the technical issues which led to the Bhopal Gas Tragedy.	CO5	K2	8.2.2
4.(a)	Describe how the EIA process of Khavda Renewable Energy Park in Gujarat will help to reduce CO2 emissions?	CO5	K3	11.3.2
4.(b)	Explain Extended Producer Responsibility. How it differs from Polluter Pay Principle?	CO5	K3	6.2.1
SECTION-B (Knowledge Level: K3)				
Attempt any one part from each question		(4 Marks x 2 Questions = 8 Marks)		
5.(a)	How do green buildings benefit the environment? Write the role of LEED and IGBC in India for green building certification.	CO4	K3	11.1.1
5.(b)	Describe the environmental degradation caused by 3D printing and write about its probable solutions.	CO4	K3	2.2.3
6.(a)	Why Environment (Protection) Act, 1986 was formulated? List the powers of the central government under EPA, 1986.	CO5	K3	6.2.1
6.(b)	What is the purpose of EIA? Explain with a flow diagram the process of EIA.	CO5	K3	11.3.2

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create